Utopian design for citizen science: collaborative thinking and writing across platforms (Workshop)

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Citizen science (CS) is not only gaining momentum across all research disciplinary fields, but it is also showing its potential to challenge the boundaries of sciences: it does so by questioning afresh the way we frame research questions, the way we collect data, the way research infrastructures are conceived and made available for research, and the way we render and communicate research within and beyond the professional research realm.

Through this workshop, we want to explore the following question: can conversations and collaborative writing about framing research questions and about research outputs in CS projects be built and circulated across several platforms - digital and physical? To answer, we will draft design concept(s) through an “utopian design” approach - critically inspired by those movements that embodied a certain understanding of technical mediation (Dorrestijn and Verbeek, 2013), in order to conceive diverse CS workflows across digital platforms and physical/analogical practices. Following the utopian design exercise, we will discuss all together about how these circulations across platforms and practices could be facilitated and stimulated to ensure that new science communication and publication formats are taken into account in the formal spaces of the open science ecosystem.

Exploring and experimenting with how to create a community of knowledge also means leveraging on experiences of how CS (Göbel, Mauermeister and Henke, 2022) and collaborative digital frameworks (Severo & Sauret, 2024) operate by creating spaces for collaborative thinking and writing. In the first parts of the workshop, the convenors will present case studies related to their research while participants will offer examples of tools and services they know.

Most CS projects applying participatory methodologies - i.e. involving non professional researchers in all the phases of the common research - happens at local scale and often aims at including vulnerable groups. This means that multilingualism in research performing and outcomes has high value in CS, that diversified formats for recording, rendering and communicating research are of high relevance, and that digital divide, digital skills and the possibility of digital tools to be embedded in analogical/physical workflows are key challenges for CS projects (Smaniotto, 2023). In particular, regarding research communication and publication formats, the challenge is not only about dissemination through social information networks, but also about being documented and archived within the current research information systems. On another hand, CS projects applying contributory methodologies heavily rely on digital technologies to implement participation (e.g. scientific crowdsourcing platforms), and AI technologies can be used. In this context, could open tools and greater interoperability give a push to involving citizens earlier on in research question development, even for contributory projects?

In this workshop, we want to address these questions and focus on two key moments of participatory CS, which are often thought of as the beginning and the end of a research process, and that both involve writing: the building of the research question and the communication about the research.

Workshop agenda:

The workshop will last 3 hours and has 3 segments. The first segment (75 mins including welcome) is dedicated to mapping ecosystems of practices between tools and humans: it will start from shorts inputs by the convenors on tools and usages of conversational and collaborative spaces within research and citizen science (approx. 50 mins), and will continue with contribution from the participants (approx. 15 mins).

We are particularly interested in exploring paths allowing us to use a variety of interconnected tools, taking into account the human practices that, beyond the digital realm, intertwine material actions, and that in many participatory CS projects are particularly central. This supports the view of the digital space as an extension of the analogical world, rather than a separate entity from it (Cardon, 2019). The concept of a path in the digital realm (Jackiewicz, 2017) encompasses a broad spectrum of scientific activities, that includes - without being limited to - documentary research and text acquisition to reading, note-taking, and sharing within a community. The tools can both proprietary and open-source and build to allow collective writing and conversation protocols (e.g. collaborative pads, pink my pad), notes taking (e.g. Obsidian, GDrive), annotation tools (Hypothes.is, Pundit), direct science communication (Hypotheses.org), CS projects facilitation (VERA). The material dimension of rendering formats mobilized in CS (e.g. fanzine, murals, mapping) will be showcased through experiences from CS projects. The attendees will then be invited to share a tool of their choice: following the number of participants, the sharing could be done through an oral pitch or pre-formatted cards that will be attached to a pre-setted board.

The second segment (45 mins) is the “utopian design” exercise itself. The attendees will be invited to welcome the diversity of the experiences and knowledge of all participants, expanding their imagination through drawing as freely as possible their utopian workflows, without feeling concerned or constrained by technological or social limitations. After an introduction to the methodology, the participants will be divided into groups (5 groups of max 6 persons each). The groups will draw paths across physical and digital conversational spaces mobilized in citizen science (and by extension in collaborative research). Each group will thus design their utopian collaborative writing and thinking ecosystem, in which a diversity of stakeholders contribute to building its blocks, and diverse collaborators intervene in thinking and writing a CS project.

The third and final segment (60 mins) will invite the 5 groups to share their designs and engage in a structured conversation on the challenges that structure the possibilities of the design workflows to be uptaken and implemented. The challenges will include: accessibility and usability - especially by vulnerable or low (digital) literacy groups, up to the capacity for them to hack the usage of the tools and services, and giving credit to the diversity of contributions - including credits taxonomies and enriched authors/contributors descriptions.

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